

allows for backward compatibility and interoperability, Wi-Fi has become very widespread and prolific in terms of the number of users and geographic capture of markets. A standard Wi-Fi device should actually work anywhere in the world as the core concept is the same, although the spectrum assignments and operation limitations are not consistent internationally.

However, most Wi-Fi networks have a limited range. A typical router at home with Wi-Fi capabilities might barely extend more than 300 feet outdoors. However, the newest IEEE standards (the ones developed after 802.11b or 802.11g) have better ranges. Wi-Fi in the 2.4 GHz spectrum has a slightly better range than the 5.0 GHz frequency block. However, depending on the antenna, the outdoor range of Wi-Fi can be improved and extended up to several kilometres.

Long range Wi-Fi is one of the most recent developments with people envisioning Wi-Fi as an alternative to satellite coverage. However, these devices are in the early stages, and have neither been certified as Wi-Fi products nor have they been certified for interoperability.

Wi-Fi security is of prime concern, especially in days of terrorist and cyber attacks

Depending on usage, range and other specifications, the power consumption varies. However, given its range and utility, the power consumption is much higher than with other low range technologies such as Bluetooth. A difficulty therefore arises when using Wi-Fi on mobile phones and other small devices with minimal battery life.

The complex nature of radio propagation at typical Wi-Fi frequencies makes it difficult to identify specific connection strengths without direct data transfer. A Wi-Fi signal only gives the signal strength for the nearest device. The reflection of these electromagnetic waves across certain media and blockage and interference are other factors that need to be considered.

As Wi-Fi networks have a limited range, jumping from one hotspot to another is a good way to permanently stay connected to the internet. The process of going around looking for Wi-Fi for wireless internet access is called Wardriving.

There are other issues to be considered such as security.